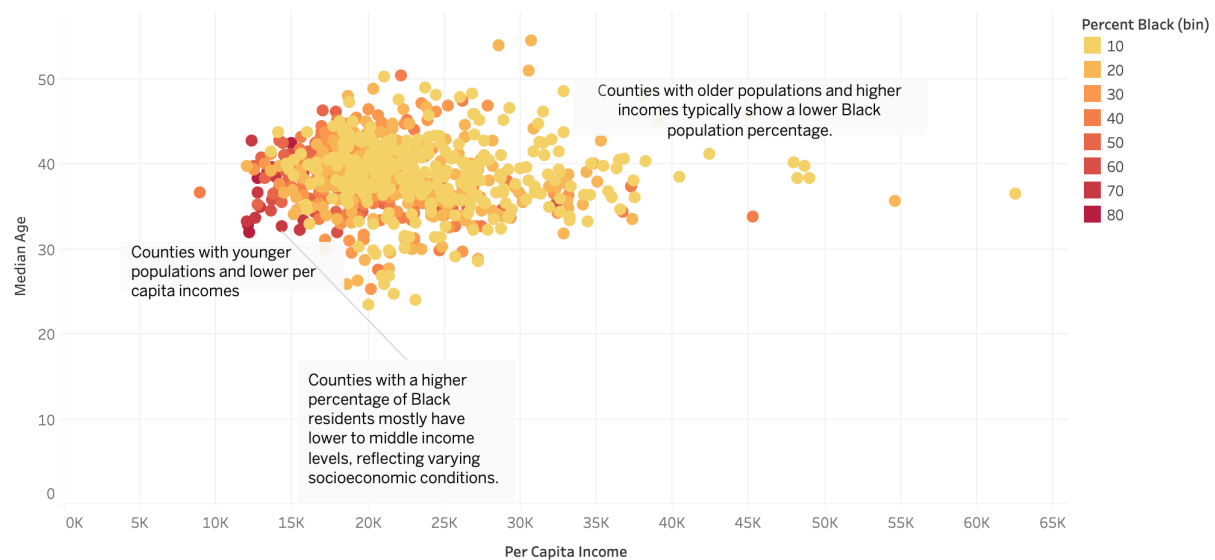


Part 1: Exploring relationships

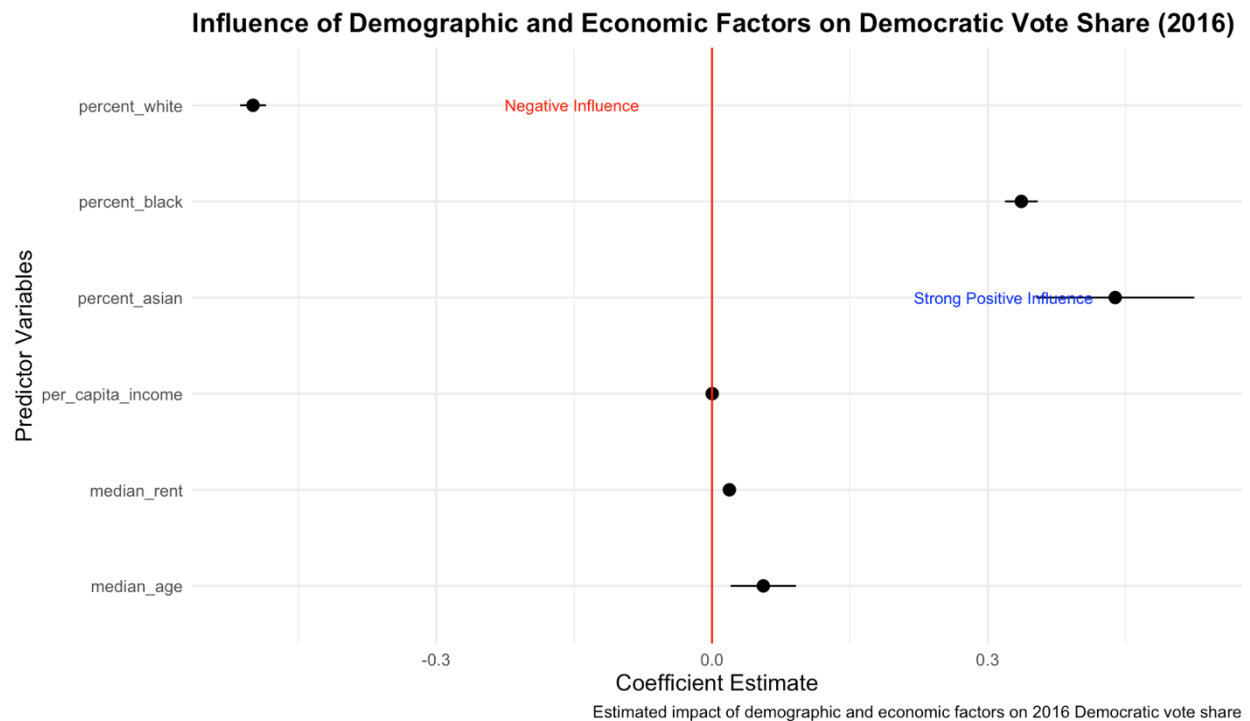
County Demographics by Age, Income, and Black Population



U.S. counties by median age and per capita income, with color indicating the percentage of Black residents.

I chose **median age**, **per capita income**, and **percentage of Black residents** to explore socioeconomic and demographic patterns across U.S. counties. The scatter plot reveals distinct clusters: counties with younger populations and lower per capita incomes tend to have higher percentages of Black residents, reflecting a concentration in lower-to-middle income areas, possibly linked to historical and systemic factors. In contrast, counties with older populations and higher incomes generally show lower percentages of Black residents. This trend could relate to patterns in urban and rural divides, economic opportunities, and shifting demographics. These insights provide context for the 2016 election, where regions with varying racial and economic compositions responded differently to candidates like Hillary Clinton, who struggled to mobilize rural and less-diverse areas, and Donald Trump, who resonated with many of these communities. The data underscores how demographic and economic factors can shape political leanings across regions.

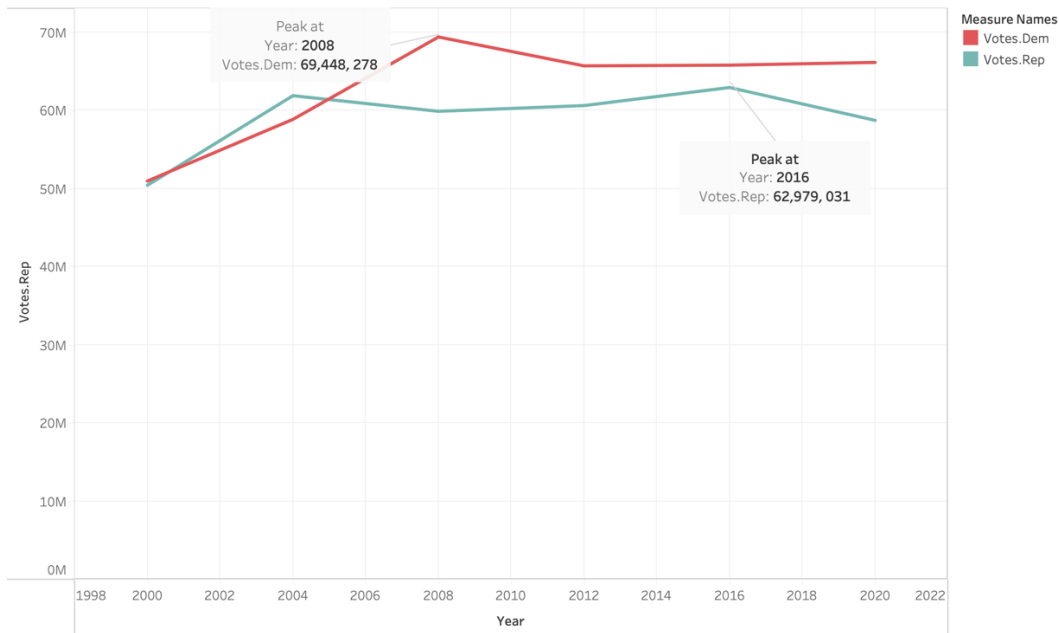
Part 2: Visualizing regression results



In analyzing the 2016 election data, I observed that certain demographic and economic factors significantly influenced Democratic vote share. Specifically, a higher percentage of White population negatively impacted the Democratic vote share, which may reflect Trump's strong appeal among predominantly White, rural areas. On the other hand, areas with higher Asian populations showed a strong positive influence on Democratic votes, which aligns with Hillary Clinton's appeal in diverse, urban regions. Factors like per capita income, median rent, and median age appear statistically insignificant in this analysis, as their confidence intervals overlap zero, suggesting they did not play a strong role in predicting Democratic support in 2016. These patterns show the demographic divides that defined the 2016 election, with Trump's success in less diverse areas contrasting with Clinton's support in more diverse and metropolitan regions.

Part 3: Time series visualization

U.S. Presidential Voting Trends: The Obama, Clinton, and Trump Impact (2000–2020)



This chart compares total votes for the Democratic (red) and Republican (teal) parties, showing growth in both with peaks around 2008 for Democratic party and 2016 for Republican Party and minor declines in later years.

In examining the voting trends from 2000 to 2020, I noticed several key shifts linked to prominent political figures. The 2008 peak in Democratic votes coincides with Barack Obama's first presidential election, reflecting a surge in enthusiasm and voter turnout, particularly among younger and minority voters drawn to his message of change. This peak tapered off slightly in subsequent elections, but Democratic support remained relatively strong. In 2016, however, there was a significant increase in Republican votes, largely driven by Donald Trump's unconventional campaign, which energized a new base and resonated with voters in traditionally Democratic regions. This shift highlights how Trump's appeal reshaped the Republican landscape, while Hillary Clinton's campaign didn't generate the same turnout among key Democratic demographics. The slight recovery in Democratic votes in 2020 suggests a reaction to Trump's presidency and the mobilization of voters for Joe Biden, who sought to build on Obama-era policies.

Part 4: Learning reflection

In exploring the 2000–2020 voting trends, I was surprised by the noticeable shifts tied to specific candidates like Obama and Trump, reflecting how individual leaders can shape party support. Modeling the 2016 election factors revealed unexpected relationships, such as the strong positive impact of the Asian population on Democratic votes and the negative influence of predominantly White areas. One of the main challenges I faced was in the first visualization, where deciding which variables to focus on and how to present them without overwhelming the viewer was a bit tricky, as was managing data inconsistencies across different election years. Additionally, creating meaningful annotations without overcrowding the chart required careful placement and choice of key points. The final scatter plot of county demographics highlighted socioeconomic divides across regions, reinforcing how factors like income, age, and race intersect in meaningful ways.